

Front-end Component Hierarchy and Roles

Notes: This is just a draft I asked chatGPT to customize based on what we have on the UI/UX proposal. Still waiting for Dominic and Vebjørn suggestions.

Root Structure

```
frontend/
├── app/
├── components/
├── features/
├── services/
├── hooks/
├── store/
├── utils/
├── types/
├── styles/
├── lib/
└── public/
```

APP ROUTES (Next.js Pages)

```
app/
├── login/
│   └── page.tsx
├── dashboard/
│   └── page.tsx
├── courses/
│   ├── [courseId]/
│   │   └── page.tsx
│   ├── assignments/
│   │   ├── [assignmentId]/
│   │   │   └── page.tsx
│   ├── coding_workspace/
│   │   ├── [submissionId]/
│   │   │   └── page.tsx
│   └── admin/
│       └── courses/
```

```
| |— sections/  
| |— monitoring/  
| |— analytics/  
| |— users/
```

FEATURES (BUSINESS LOGIC LAYER)

features/

```
| |— auth/  
| |   |— login.service.ts  
| |   |— auth.api.ts  
  
| |— submissions/  
| |   |— submission.service.ts  
| |   |— grading.engine.ts  
  
| |— assignments/  
| |   |— assignment.service.ts  
  
| |— upload/  
| |   |— zipParser.ts  
| |   |— upload.service.ts  
  
| |— analytics/  
| |   |— stats.service.ts
```

COMPONENTS (UI LAYER)

components/

```
| |— ui/  
| |   |— Button.tsx  
| |   |— Card.tsx  
| |   |— Modal.tsx  
  
| |— editor/  
| |   |— CodeEditor.tsx  
  
| |— upload/  
| |   |— ZipUploader.tsx  
  
| |— test-results/  
| |   |— TestCasePanel.tsx  
  
| |— feedback/
```

| |— FeedbackPanel.tsx

|— terminal/
| |— Terminal.tsx

|— grading/
| |— GradeSummary.tsx

|— layout/
| |— Sidebar.tsx
| |— TopBar.tsx

SERVICES (API LAYER)

services/
|— apiClient.ts
|— auth.ts
|— submissions.ts
|— assignments.ts
|— admin.ts

Types

types/
|— user.ts
|— assignment.ts
|— submission.ts
|— testCase.ts
|— grading.ts

Utils

utils/
|— zipValidator.ts
|— fileParser.ts
|— scoreCalculator.ts
|— formatTime.ts

Roles:

Student:

- See the courses that they are enrolled in for the current semester (grid and list view)
- Submits code (zip bulk upload)
- Submit (5 uploads/hour)
- View rubric
- View constraints
- View feedback and score
- See the history/score of each attempt before closing the section (*In discussion whether to keep this feature or not*)

Admin/Instructor:

- Manage courses
- Monitoring (token limit, system limits, upload limit)
- User management tools (assign instructor, TA)
- Create courses
- Create sections
- Deactivate courses
- Deactivate sections
- Create assignments
- CRUD rubrics
- CRUD constraints
- Set a deadline for assignments
- Add/edit test cases
- Override the autograding (future milestone)

TA:

- View assigned courses and sections
- View only (rubric, constraints, deadline) of each assignment
- Run test cases
- Override the autograding (future milestone)